

Livestream 058.2: "From pixels to planning: scale free active inference"

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hello and welcome everyone it is November 13th 2024 we're live and kicking off the fourth applied active inference Symposium it will be a great program and we will begin with Carl Friston and Lance dcosta discussing scale free active inference we discussed in live stream 58.1 one with Lance last week and in 58.0 with Aron two weeks ago a bit about this work and today Carl thank you so much for joining and presenting on it so looking forward to it and throughout in the live chat please feel free to write any comments thanks again everyone for joining and looking forward to this well thank you Daniel it's a great pleasure to H kick off the Symposium this year um I'm going to rehearse um a presentation I made at the um active inference Workshop um a few months ago but this time I hope in a slightly more relaxed uh and unpack way so I'm uh fondly hoping for lots of questions and uh embellishments from the uh from the audience and uh and from Lance um this is a bit of a colloquial presentation it's just going to focus on something specifically that we've been working on for the past few months which is effectively addressing the question how can one scale active inference for high dimensional real world problems and the solution that we've been pursuing is to adopt a scale-free approach um this work has been written up um and put on archives um however it's also been submitted for uh publication peer reviewed publication to a special collection um in uh memory of Herman hen um and we've equipped the peer reviewed version with a little forwards I just wanted to frame or preempt the scientific part of this presentation just by motivating why why the work of Herman hen is foundational and under writes many of the um ideas and procedures that underly this scale-free approach to active inference so Herman hen was famous for synergetics um a physics Le approach to self-organization um and in particular the importance of understanding the coupling between different scales of Dynamics in self-organization and if you go to the Wikipedia entry you will read about macroscopic and microscopic variables and how they are coupled to each other in a in the fashion of circular causality um so there's both bottom up and top down causation where the top down causation

is sometimes read in terms of um the slaving principle I'm going to present this approach um to harnessing this um notion of a scale-free universe uh in terms of installing that scale free aspect in the generative models that we bring to the table to solve decision- making to uh simulate or um emulate active inference under discrete State space models so let me start at the beginning um the nature of things and the free energy principle um and this is just to rehearse the fundamentals of the free energy principle it all rests upon the notion of a Markoff blanket where we can separate internal and external states by boundary States or states that constitute the mark of blanket namely the sensory States and the active States and for those people not familiar with um this particular partition of states of any random dynamical system it's just specifying systemically um a system say a brain in terms of its inputs and outputs and um just formalizing the conditional dependencies that Define what is an input to a brain or an agent and what is its output in terms of the sensory and active states that thereby render the system of interest in this instance a brain open to the environment the econ Nish um the heat bath if you're a physicist or chemist um in virtue of the fact that the inside can influence the outside via the um active States the outputs and the outside can influence inside via the sensory States or the um or the inputs and with this setup we can elaborate um a kind of physics namely uh um basing mechanics that is almost well formally um related um and inherits exactly the same mathematical and functional form as all the other mechanics that we find in physics including quantum mechanics sta statistical mechanics and classical mechanics the the only thing that separates the um the basy mechanics is the fact that we have in mind explicitly this partition between internal States and external States and the uh blanket states that intervene and couple the internal to the external and that basically mechanics is is effectively summarized here in terms of um variational and expected free energy um where we can read the expected free energy as effectively PRI beliefs about the kind of outputs or actions that um in this instance a brain could exert on the external States um and the variational free energy keeping implicitly the um the internal States a a good model so it's tracking in a probalistic sense the um the Dynamics on the outside and we can combine this to uh create a a generalized free energy um that now te logically or or perhaps even an anthropomorphically we can now read the gradient flows that subtend this phasing mechanics in terms of action and perception just by identifying the internal and active states of any agent as autonomous States and noting that they effectively um um flow on the same free energy functionals so that's the basic setup um but what I want to do before um applying the free principle in the service of things like active inference is just back up and reflect upon the

fundamental role of the Markoff blanket and how that leads to a scale invariant picture or view of Dynamics and self organization so what I'm going to assume is that ultimately we will assume that the basic mechanics means that the Dynamics of anything that is defined in terms of its boundary or Markoff blanket it can be described in terms of active inference but before that we have to think about the nature of things and in particular if one goes back to the particular physics paper the nature of the states that constitute things or particles and there's a bit of a subtlety here because before you can start talking about States you really have to Define what is a state is it the state of something is it the state of a particle and that presents an issue because you're defining a partition in terms of this particular partition into internal external and Boundary or blanket States so there's a problem there's a slight topological problem but there's a slight problem in resolving how you define states of things or particles and particles that possess States and the resolution of that is to adopt a recursive definition so that the particles constitute a set of microscopic blanket and internal States while the states are a set of macroscopic eigen functions eigen modes or just simply mixtures of the blanket States so I'm here deliberately using the notion of microscopic and macroscopic States just in order as a nod to synergetics and particular kind of synergetics that Herman Haken developed and subsequently championed so this is cartooned or illustrated on the right hand side here so we sort of take some states and then we apply a particular partition on the basis of which states are coupled to which other states and the conditional independences that ensue and then having done that we can then take mixtures of the boundary states that then generate new States and then we can effectively by taking those mixtures we effectively reduce the dimensionality so that the higher scale coarse grains the lower scale and then we reduce it and then we just rinse wash and repeat and we do that and every time we sort of pass through this cycle of RG operators reduction and coarse graining operators we move to one scale above successively and this is the scale invariant aspect of that I want to leverage in the remainder of this presentation I use the word RG deliberately because one way of looking at this or articulating this is to use the apparatus of the renormalization group so this is where renormalization gets into the game that you're recursively applying these operators Coarse graining things in a way that conserves their Dynamics and that's an easy criterion to meet for us because we know that anything that has a particular partition can be described in terms of a gradient flow on variation or generalized free energy so we have everything that we require in order to apply the apparatus of the renormalization

group so that what that basically does for us once we recognize that particles are composed of states where the states are ion mixtures of particles in this scale invariant fashion gives us a view of any universe that comprises things of things of things um as um inherently uh resolving a chicken and egg problem through scal invariance by application of the renormalization group so here's just another illustration of how that actually works in practice when you write down the equations I'm not going to bother going through the equations this is just used iconically um just to make the observation that as you move up scales um so for example in an evolutionary context this would be the sort of the Dynamics of phenotypes and then we move up to the an evolutionary scale by um effectively Co graining in a way that Herman hen um would exactly prescribe in terms of taking those slow macroscopic variables that decay very very slowly um which are technically the igen mixtures equipped with very small um igen values whose real part approaches zero From Below effectively saying that these patterns Decay very very very slowly and they have a kind of persistence that lends higher and higher scales a temporal dilation so things change more and more slowly as you increasingly uh cor grain things and that this falls out of the maths um when you simply apply a particular partition and put that partition I in um as the um the course grading operator in this uh in these in this scale free formalism so in short basy mechanics entail a generative model of external Dynamics but we've just seen that just by defining things in terms of particles that have States and defining States in terms of the ion functions of particles that external Dynamics are inevitably scal inv variant it can be no other way and therefore the generative model must be scale invariant or when expressed as a graphical model scale free and I just slip that in as a um a nod to the fact that people often talk about scale freeness um generally speaking when people apply the notion of scale in variance to anything if that thing is a graph then you can call it scale free so a scale free thing is a um is a graph that has has this scal inv variance crucially in both time and in space because we are dealing with dynamical systems and we're doing this um Dimension reduction or course graining um using these um migen functions that necessarily pick out these slow macroscopic sometimes called stable modes they're stable because they they don't Decay or dis ipate um almost infinitely quickly so they're stable in the sense that they Decay very very slowly so how can we harness that observation that truism at least truism under the world as described by hen um in um the construction of plausible generative models and in particular um generalized Markoff decision or partially observed Markoff decision processes um well what we can do is we can just look at the different kinds of depth um and ask at what point would we um be able to leverage um

this renormalizing aspect of the cause effect architecture of Any Given world that we're trying to navigate and predict so here's uh the normal um depiction of a um a generalized um Mark of decision process and by generalized what I mean is that there's um an explicit representation of not just the states of things that are generating outputs or things that can be measured or observed on the um on the sensory sector of a marov blanket but also their Dynamics in terms of the paths that um specify the transitions between state as we know normally encoded in terms of a transition tensor be here um so just by putting in the paths as explicit random variables that accompany for each given Factor um the hidden States we have a generalized kind of uh Markoff decision process uh encoded in the usual way in this instance um in terms of tensors um that themselves can be Ariz to distributions so um that parameterization equips these kinds of models with a parametric depth in the sense that if something is deep parametrically that means that the parameters of various probability distributions are themselves random variables that have their own probability distributions so for example the likelihood mapping that maps from sensory states to outcomes here uh usually encoded by um an a tensor here mapping from latent states to observations is itself encoded in terms of um diset parameters um blending this uh kind of model A parametric depth and um one can ask what the implications of that are well the implications are another kind of depth in terms of um you now have to optimize or the B apply basic mechanics to the latent States the hidden state and that would be inference and then you also have to um consider at a slightly slow time scale the optimization of the derish parameters and that would correspond to learning and so on um if I wanted to go to the structural learning of the model in of it in and of itself we also have um implicitly a temporal depth in the sense that we're going to be raring out into the future and that there are dynamics here um and this Temple depth is generalized in in virtue of having um these explicit uh path variables that we're going to uh exploit later on we can also have hierarchical depth in the sense that we can put um compose many markof decision processes um um in the in the sense that the outputs of one process now constitute the inputs or the empirical prior the top- down constraints the induct biases on the Dynamics and states of the level below and the Dynamics are um um the kind of inputs or sorry outputs of the uh upper level that provide empirical PL on the paths and the initial conditions are supplied or mediated by a mapping between higher levels slower States um and the initial conditions or initial States of the lower uh of the lower level by this D tensor here that you can lump together um in terms of sort of high you can lump together the D and the E tensors they play the role effectively of the a tensors in sort of um mapping from one level to the uh the

next level there's also factorial depth which many people will be familiar with in terms of um for any one of these um lumped States here there may be many many different factors that have an a certain Independence structure such that you have interactions between factors in determining outcomes um you can look at this in terms of uh agative model when equipped with factorial depth effectively entangles latent states to generate um outputs or outcomes which themselves can be prior constraints on the states and dynamics of the of the level below such that the inversion of these models effectively disentangles the data the content that is being observed to explain it um in terms of a disentangled representation which is the um the latent State and dynamics of the generative model so what we have here then is um a generalized discrete State space model with Paths of as random variables um just to um return to this um fundamental aspect that as one ascends scales in real life in um you in terms of uh the external States things slow down in virtue of the um the the the the synergetic um mapping from Fast microscopic States uh to um slow macroscopic States um the same thing can now be implemented in these um markof decision processes that have a hierarchical depth um and the way that is done is effectively to discretize or quantize time such that there there are more updates at any given level than any um than the level Above So for every um update at this level there are uh say three updates or two updates at the level below and that um affords a temporal scaling um in accord with the scaling variance um implied by the synergetic view of self-organization I've tried to cartoon that here in terms of if you imagine lots of these um little markof decision processes populating this uh circumference here and that as we go deeper and deeper and deeper into a model with hierarchical depth as we consider updates in Universal or clock time there are more updates at the peripheral level exposed say to the the um external states of entire brain um relative to the relative to the updates as you get deeper and deeper and deeper in into the model so where the temporal depth um becomes if you like intimately tied to the hierarchical depth the higher the hierarchy the slower the updates which is why um we generally parameterize the initial conditions um or generate the initial conditions and the initial path from the states above and that path can pursue um um a trajectory over multiple time steps until a certain um Horizon uh and which case it then sends messages back to the level above likelihood messages um that then cause belief updating um in the subsequent Step at the level above and then generating priors about the subsequent path the next PATH at the level below and then so on recursively uh to the depth of the model um so that's the time bit what about the spatial bit what about the spatial core graining the lumping together of various States um and of course this depends upon the particular partition so in the General Physics

formulation you'd be looking at uh Markoff blankets and markof blankets in Practical applications one can take a shortcut um and that shortcut effectively rests upon um the simplifying assumption that for most systems that are um composed of um coupled um coupled uh dynamical systems um the interactions are usually local so if we um just think about um some um um system that um can be described in terms of local interactions so for example um in a um a ukian world um that comprises some ma massive objects that bounce around and if we ignore gravity for the moment then they're only going to influence each other or depend upon each other when they're touching when they're um when they're proximate um effectively what that means is the the the mark the particular partition that defines all the Markoff blankets in um in this particular State space um um is composed of lots of local little tiles um so what we can do is we can use a um a device um very prevalent or indeed ubiquitous and applications of the renormalization group which is called a uh a block or a spin operator that just groups together local tiles or quadrants of a say two-dimensional arrangement of time of of States so I've Illustrated that here just by um in terms of um How We Do the spatial cor graining in these um what will refer to as a renormalizing generative uh model with spin block uh Transformations um in terms of um this little um um this hidden state or hidden factor with its states paths here um is responsible for generating predictions about the Dynamics of these two um factors um at the um at the lower level and in so doing what we are effectively doing is successively grouping together sets of sets of sets of sets so that any set or group at one level is accountable for or trying to generate the Dynamics of a small local group at the at the lower level um this is quite fundamental this kind of Architecture is quite fundamental because what that means is anything at the lower level only has one parent and if it only has one parent and if you remember the markof blanket is constituted by the the the um the parent and the children the parents of the children because there are no co-parents there are no parents of the children and what this means practically is that we can now ignore dependencies between or effectively resolve or dissolve any dependences between the factors at any given level because each factor is only responsible for its children and it doesn't share any children in virtue of this um particular spin block transformation so this leads to um a very efficient kind of generative model um well certainly an efficient kind of inversion of the gened model um where we've got all these converging streams but the streams don't have to um uh talk to each other until at the point that they come together through these blocking or spin block um Transformations so at some point all the tiles or the groups at the lowest level will actually come together in um uh at the top or very deep in

the model at which point we're now encoding quite long trajectories technically 2^D to the 2^D of the depth minus one in terms of updates but at the lower level there are no interactions between the different blocks as we Ascend them and this is just to remind you we can treat these D and E the initial States and the initial paths effectively as likelihoods that couple one scale or level to the scale below so how uh with if that's the kind of model that we aspire to because we think that's appropriate for any world that shows this scale invariant aspect in space and in time or in state space and in time how are we going to build one of these things and this is the sort of second if you like thing that we have been working on because it's going to be very difficult to specify by hand these scale-free or renormalizing generative models we really want the model to learn itself so basically how does one build a renormalizing generative model and what we're going to do we're going to use very fast structure learning and this particular kind of structure learning is necessarily recursive and uh in the sense that we're going to build these models through the recursive application of these RG operators or renormalizing operators so what's fast structure learning so fast structure learning basically equips the model with unique states and transitions as they are encountered so to put that simply what we're going to do is grow a model from scratch from nothing and how are we're going to do that well we're effectively going to take one observation and say okay that was generated by some latent State and then we're going to take the next observation and say okay I haven't seen that particular observation before so I'm going to induce a second latent State and furthermore I've seen these two of observations in sequence therefore I'm going to encode that with a transition from the first to the second state and then I just repeat I take the the third one and if I haven't seen that before I induce a third latent State and a third a second transition accumulating all the Unseen or unique instances or of observations in a likelihood M mapping that just grows and grows and grows until we have seen something before so we don't induce a new state or implicitly column of the likelihood Matrix if we've seen uh the previous state that's just a um an instance of something that has been seen before um it may well now come along with a different uh path so the next state may not be the same as the the subsequent State following the previously scene instance of this state and therefore I'm going to induce A New Path and slowly grow my B tensors and my probability transition matrices until I can um basically encode all the dynamical structure of the effectively the training sequence of observations summarized in a a maximally efficient way and I mean maximally efficient in a technical sense um in the sense that the implicit map that we are

growing from from scratch μ has the highest Mutual information between the latent States and the observations that are at hand μ the way that you can think of that in terms of μ free energies that the expected free energy μ is just the mutual information in the absence of any expected cost so the expected free energy is just the mutual information of any mapping μ μ μ plus μ or minus an expected cost where cost is defined the usual way in terms of constraints or prior preferences so one can actually look at this uh fast structure learning μ as a particular kind of active inference but not about μ it's not active inference Over States or Active Learning over parameters it's active selection of the right model in terms of the size of the likelihood tensors μ and the transition uh tensors μ and the criteria for whether we accept a new latent state or a new transition is simply does it optimize or improve the expected free energy namely does it preserve or conserve or maximize the the the mutual information so we can actually look at this structure learning as just another instance of act inference but here deployed in terms of μ μ in terms of basic model selection or some some people call this uh structure learning that has exactly the same rules as μ planning μ and μ uh and μ accepting μ parameter updates provided they they maximize expected free energy as well μ the application of this kind of uh fast structure learning to μ these renormalizing models U looks complicated here's the algorithmic description of it μ again please don't worry about the details here it it's relatively simple you know all we're doing basically is ingesting μ observations but we're just doing it for local groups μ and then we are combining the initial conditions and the paths into new outcomes for the next level μ and so because we are doing this for local groups the number of groups just get smaller and smaller and smaller as you as you get higher and higher and higher but obviously the number of combinations of paths and state initial States increases as you get deeper and deeper into the model and you may be asking well you know how do you ban that you know how can you μ possibly assimilate all the data without having an extremely large number of latent States very deep in the model that are now encoding effectively evolving patterns or paths over quite a large number of time points into the future well it's trivial to do that because the size of the highest level I'll call them generalized states that basically generate paths and initial conditions for the lower level μ is upper bounded by the number of elements in your training set μ so that you can control and upperbound the size of these renormalizing operators or likelihood mappings and transition μ μ tensors simply by giving it canonical training data the kind of data that you want this μ agent to generate or this μ μ agent to recognize and thereby predict μ so here's a a particular example of that μ what we've done here is just ask can we very very

quickly learn a generative model that will generate a um a simple movie and this movie is simple because um it begins and ends in the same state so it just Loops over and over and over again um so we only have to supply one Loop um to this fast structure learning so that the model builds itself and can then be used in generative mode to generate the movie time and time and time again um so here's one depiction of um the results of the learning the the F structure learning um and subsequent inference when exposed to um this kind of image so the coarse graining here at the very bottom in in fact um uses um um a slightly finessed version of the spin block Operator by um harvesting um local igen modes or igen ion images um of little patches of of the video to get it into a reduced or coarse grained representation that then is just passed up using these spin block operators and in this in we just need uh two levels to the model to uh have the highest level see the entire image and then if we present the image um um after discretization using the singular decomposition or the sequences of images to a model it learns the model and then we can generate um we can generate um little movies simply by moving through these paths or episodes I use the word episode um because remember these generalized States the highest level encode paths or trajectories into over um 2 to the N minus one time steps in the future when n is the um is is the uh the depth of of the model and these episodes themselves have Dynamics so we just go through a little Loop here go back to the beginning and thereby generate um this image at the bottom what else can we do with these kinds of models well one thing we can do is um present not the entire data but just a little prompt um so in this example what we've done is actually show it just the upper right quadrant of the movie and the and see what this um and this generative model um believes is going on in terms of the um the predictive post area in output space in image space in this instance now because during training it's only learn to recognize a dove flapping her wings um that's all it can believe can is the cause of its Sensations even if those Sensations are very partial so basically what we have here is a kind of completion that inherits from the fact that this generative model only knows about latent causes that constitute an entire Dove flapping her wings so um even though this is what the dove is actually um exposed to it's sensing this is the predicted after a couple of frames it very quickly um is able to predict um what it would what it should be seeing or what it would have seen had it been given the entire uh data and another way of thinking about this um remarkable ability to sort fill in the gaps um is from the point of view of compression so just a little digression here um a lot of the basy mechanics and um variation inference that uh ensues from variational mechanics can also be likened to um the to um the same fundamentals underwrite um um efficient information transfer so we've talked

about maximizing Mutual information in the context of uh um algorithmic complexity and um the associated Universal computation um then what the objective function looks like is basically how much can I compress this content um to represent it in the most efficient way so you can look at this past structure learning under these renormalizing or scale-free generative models as the most efficient way of compressing whatever your data you've been exposed to um and because we've compressed it in such a way that it can only now recognize a dove flapping her wings um then it can if you like fill in the gaps in a very efficient way when it's given partial or noisy data and I use the word noisy here because you can regard the missing three quadrants here as extremely imprecise data data that has extremely low signal to noise for example so um here's another example in the previous example we we um took a um effectively uh a movie or a dynamic or um a system that had a periodic orbit um um which was each flap of the Wings will this work for um a periodic orbits and and generally for um stochastic um uh chaos um yes it works it works relatively efficiently so this example um we um used training data that was generated um um through creating images of a white ball that moved with a chaotic trajectory governed by a uh a loren system so this is the kind of system that governs or is used to describe chaotic events um in meteorological systems or in fluid convection um so by applying exactly the same procedures as in Pre in the previous example um we now uh generate this instance a three-level um um hierarchal or renormalizing model um that now accommodates the um stochastic or Randomness induced by not only random fluctuations on the Dynamics of this stochastic uh random dynamical system uh but also if there were any deterministic aspects the deterministic uh um um chaos that ensues from exponential d convergence of trajectories that now has been summarized stochastically in terms of probalistic um uh transition matrices here which means that um we can having compressed the training data we can now withhold not a quadrant but just simply withhold the data to stop the data uh but the model will keep on generating um what it predicts would have happened given the statistics of the Dynamics that have been in store in this renormalizing generative model and what we've done here is um um for the first um few hundred time bins at the lowest level we basically rendered the images or the input very very imprecise so that everything is governed now or everything is generated from this top- down Prize or posterior um um posterior um predicted posteriors that inherit from from precise PRI is because there's no precise likelihood um and then we've actually uh removed the U um removed the um imprecise input completely and just allowed the um the agent to generate its own beliefs about what's about what's going on um so here this is fairly precise in the first instance and then we

remove completely the um stimulus but it still uh imagines or generates its or predicts um what's going on in terms of this chaotic um Dynamics um and then um with a degree of uncertainty then um continues to then generate um the um the Dynamics associated with this kind of stochastic chaos here's a pretty example um using exactly the same uh technology and approach um but here explicitly um talking about um video compression um so we literally just presented a video of naturalistic Dynamics in this instance a little bird a robin um feeding um itself um and um compress that using these uh this renormalizing um generative model um such that we can again simply remove stimuli at certain points um in the video um and yet the the in in the agent's mind nothing has really happened because all it knows is that if it was like this um previously then this must have happened um and so on through this in this instance um orbit that doesn't have a a closure it's just a trajectory through from the beginning to the end of the sequence and here's the postor prediction after just being exposed to a couple of frames right at the beginning um in the first um six uh time frames of um of about 12 sorry of um 18 um this example moves applies exactly the same um approach um as previously but now we're not talking about um ingesting or learning or compressing images we're talking now about compressing or ingesting sound files uh um and that is if you like a simple or limiting case of this two-dimensional uh compression that we talked about before so in this instance now what we're doing is compressing one-dimensional frequency summaries of a particular um wave file or a um file um that um describes fluctuations and the intensity of uh of a sound that then is um summarized into terms of a continuous wavelet transform um and then the succession of different trajectories in this um time frequency space is then ingested or assimilated or learned by the generative model um now unfortunately you can't hear this um so I'm not going to bother playing it but um it um this is what the agent was trained on it's basically J piano um and these are the um ensuing predictions in the absence of any inputs it's been used to generate music um using um the stochastic transitions um through various bars of Music um that it has learned um returning to one or other bars of the munic of the music when it reaches uh the end of the orbit to generate something that um sounds at least Jazzy if not oh no started it but you can't hear that so I'm not going to pursue that so to summarize what we've done so far um we've gone beyond generative AI in the sense of um um having a a true generated model um uh that entails active inference in agency um and in the setting of these um well sorry what we are going to do now is go beyond just simply generating uh content um having been uh trained on the basis of some training data um and we're going to now consider um agency by putting action into the mix um and in the setting of um the simplified setting of optimal State action

policies where we can effectively um ignore the imperative to maximize expected Information Gain or um um explore in the sense of uh responding to epistemic affordances um we can um do one of two things we can either use some Universal function approximator say deep learning to optimize the mapping from sensory to active states to maximize the um expected cost or the expected utility part the expected free energy um or we can use active inference to realize predictions under enormous izing generative model of rewarded events so what I'm doing here is deliberately setting up a dichotomy that tries to distinguish the active inference approach to purposeful Behavior where the purpose is provided purely by the prior preferences or the um um or the the uh the constraints usually encoded in the c um uh tensor um against a sort of reinforcement learning approach and this is rather busy slide but um I think it's probably worth just um uh briefly uh rehearsing the distinctions between active inflence and reinforcement learning so the you know the story with respect to action as a gradient flow on free energy which basically reduces to um picking those actions that maximize the predictive accuracy um in the sense that if you look at the expression for the variation free energy the only thing that you can change um is the accuracy in the sense that free energy can be written as accuracy minus complexity minus accuracy and the complexity um is does not depend upon action whereas the accuracy does so that picture basically says um I can describe behavior in terms of a generative model that um is optimized through um perception uh in the usual way um cast in terms of say variational inference um that basically generates predictions about what should happen next and then I can pick my actions to fulfill those predictions in continuous state for formulations this would be without minimizing the prediction error usually the proceptive domain but you can generalize this to any any consequence of action I'm just going to pick that action that brings about or is most likely to bring about what I predict will happen next and what I predict will happen next um is optimized through perception through variational inference based upon my learned and selected um generative model so this is very similar to sort of um model predictive control um and neurobiology could be cast in terms of um um ual control theory um which puts action in the game basically of realizing evidencing um predicted States so I've tried to summarize that here just in terms of um you know our Markoff blanket where we're optimizing our beliefs about the world um noting that planning in this instance so um planning as inference uh is all about the path it does not have to be about action it's just about how does this world unfold um and from the point of view of these scale-free generative models we're really thinking about how do episodes follow each other which episode is going to follow this going to um um this

episode and we can say well you know you are in control of that um you know imagine a um a benevolent World in which everything happens according to um your beliefs about how to be an expert in this particular world and the agent can plan a series of episodes generating predictions all the way down to the bottom which then slavishly action can fulfill just by fulfilling those predictions um so one could in some sense uh think of this in terms of controller's inference at the bottom level in the spirit of mod predictive control and planning as inference on the top level so belt and braces in terms of realizing active inference in this particular example where I repeat um we are really just worried about um State action policies um why are we worried about State action policies well these are the kind of policies that reward learning schemes um are aimed at um and have a similar architecture um but um to my mind a slightly more um um engineered architecture first of all you actually need explicit inputs which are rewards um there are some proxy for um the constraints uh you know um um under which um the basic mechanics would normally operate um but more importantly you're you're basically sending um your inference scheme is basically about selecting the right policy I've used Q learning here um um to maximize the disc counted reward um so similarities and differences but we're going to focus on uh active inference um uh in this in the particular setting of these renormalizing models where repeat planning and thinking or imagining a future can proceed at the very highest level that has no notion of action they're just patterns that unfold they may or not be caused by me I don't know I don't care um but the action at the lowest level the refle Es as it were um do um are in a position to make my um post predictive posters come true simply by picking those actions that realize the predicted outcomes so how are we going to leverage that kind of merely reflexive active inference um in the context of building or learning from scratch um models that would be apt for doing something implementing a state action policy um and I'm going to um appeal here again to this notion of um active selection as one way of um looking at um action which can always be at least phys mathematically expressed in terms of um of um doing something if it minimizes expected free energy um and in this instance um what we can do is actually apply it not to the selection of the model but the selection of the training data so if we just um apply the the dictat that you only select something if it um um minimizes expected free energy the includes um the expected cost what that basically means is I'm only going to select those data features are those training sequences that don't come along with a high cost namely those that are rewarded um and that's what we're going to do here we're basically going to um learn to play um a very simplified um game of pong where this ball bounces around inside a circle um and

we can move the bat um in a way to uh return the ball so that we can avoid penalties when the ball hits the upper boundary um and secure a reward um whenever um those contact of the ball and the bat and how can we learn this kind of dynamic these episodes in exactly the same way we learned the video of the robin or the the um the Dove flapping her wings well what we can do is we can just um select um instances of random play that conform with um our um dictat that we cannot um include those costly events and so what we've done here is just generate training data training sequences using random play but om anything that does not that contains a costly outcome and furthermore just to make things more efficient we've restarted the random play after a reward is secured and then just use the subsequent play if it actually attained a subsequent reward so basically chaining together bootstrapping together or splicing together sequences of rewarded uh play um and then use that to um uh to um submitting that for fast structure learning and now we have a generative model of expert play effectively or a generative model that has that entails a state action policy that can be realized through um um reflexive active influence so active data selection um and I'm um sort of cartooning that here in terms of applying a certain kind of Maxwell's demon that only lets through non-costly episodes um basically yes admit the sequence for fast structure learning if there is a reward if not ignore this ignore this sequence um and this effectively learns um or furnishes a generative model events that lead to a reward and nothing else so now this agent can only recognize expert play and therefore all of its predictions have to be uh the predictions of an expert player and because action is fulfilling the predictions it will look as if it's an it is an expert player and here are the um the Learned trajectories um at the highest level of different episodes that we'll see uh um shown in in an alternative format uh in the next slide so here's the initial starting episodes little events here uh the red dots correspond to reward so you know first hit second hit and then it falls into a nice little orbit and this comp completes um this orbit indefinitely by in it's the way that it has learned so remember it's only been trained on random data so this particular style of play is unique to this agent but it is sufficient now to play with 100% um um uh performance um simply because it's now going round and round its orbit very much like the wing flapping her dug flapping her wings um these are just different ways of portraying the same um the same Dynamics and the same the things that have been learned um the first eight frames and then subsequent frames um of expert play um this is the usual format showing the posteriors at different levels the posteriors over the states and the posteriors or predicted posteriors over the paths and then rewarded players are function of the uh free energy at different levels here illustrated by the by the red dots every time there's a hit up until

512 time frames here and here's this rather long orbit through 50 episodes um each four uh time steps in in length that subtends this kind of play this is an interesting one and speaks to the fact that we've actually switched on inductive inference at the highest level um so what does that mean well what we've done is just um um basically take each of these highest level States worked out the sequence that it encodes as by um um Computing or projecting down through the model and if that sequence includes a hit um we said that's an intended State that's the kind of state that I want to work towards through the successive um four um time step epochs um and if I want to be at some point in the future in one of these intended States in this instance rewarded States or episodes um what states must I avoid in order to preserve or conserve the possibility of getting in or getting to one of these intended States um basically that's the technology behind inducting inference and it's sort of summarized here in terms of the latent States of uh you know 50 or so of these latent States in terms of how many time steps would it would it what's the shortest number of time steps until the intended State um and basically um we um as we move from state to state to state we want to keep in the white areas here avoiding these dead ends if you like from which there is no possib of reaching an intended State and it's really trivial to evaluate these um areas that you have to avoid because you know the probability transition matrices so you just multiply it by itself a given number of times and you can work out can I get from this state to this state and if you can't um then you get this um black area here and you know you have to move from here to here to here to here to get to um your um your intended state so that's inductive inference in brief and we've applied at the top level so we just the you just told the agent this world um is under your control imagine you know imagine what you want um noting that these particular episodes or events contain you are things that you should pass through or should experience um and then deploying inductive inference to basically predict what will happen all the way down at the bottom and then using reflexes to make that happen um we can apply exactly the same um technology or the same procedure to different kinds of games that a slightly more um complicated game um me to emulate breakout um where we've introduced uh certain stochasticity in terms of the initial conditions and uh sticky action but the the overall behavior is the same it's just basically choosing paths through episodes into the future given what knows that can be achieved in order to get to uh rewarded States so um effectively what we've done is create um an attracting set of episodes at the highest level of these scale-free um um generative models where all the allowable paths and paths now in this generalized State space or event space lead to reward um we can go a little bit further and just um add in uh various

episodes by merging uh or assimilating new data new training data if and only if um the path actually leads to an existing um part of this attracting set and I've just illustrated that here um in this final example so this is the path or the orbit or the attracting set that was learned at the highest level by the um the ping pong playing uh agent but now we can um fill in um fill in this um effectively um allowable Dynamics to include all states that lead to a path on the orbit um and or thereby all paths will eventually lead to reward or to roam so we can augment the model of expert play with paths that lead to an orbit namely this attracting set um and paths that lead to event that lead to the orbit um recursively and get quite a dense set of allowable transitions that we can be guaranteed always keep within the attracting set that will contain these preferred States or um states that we have um constraints uh where um we have constrained those states that are not visited simply because they cannot be recognized and that's it thank you very much indeed amazing Carl thank you awesome okay Aron if you'd like to lead in with a first question and Lance too thank you but go for R first uh yeah thanks very much Carl um so my first question is um based on a sentence in the paper which talks about any model being sparse must be renormalizable could you say a little bit more about why that must be oh I can't remember give me a clue uh I I'm presuming it has to do with Markoff blankets and yes yes yeah that's as far as I got I see yes no I I remember now um I it may have been in reference to some work done by Dalton um showing that with probability one provided the system is large enough there will be markoff blankets and as soon as there are Markoff blankets there is now an opportunity to take Markoff blankets or Markoff blankets in the spirit of um that applying the renormalization group that that I started off with um so as soon as you've got um as as soon as any um sparsely coupled random dynamical system um can now support a um a particular partition into Markoff blankets you've now got the opportunity to take Markoff blankets and Markoff blankets and then apply that renormalizing procedure that I said before so all you need to posit is any system um that has a Markoff blanket at any given level and as soon as you've got that you've got Markoff blankets all the way up um not necessarily all the way down but um you certainly got them all the way up um the the physics approach would would actually um also actually um require for for there to be markup blankets all the way down as well but that's that's that's another issue um so um what that basically means is that um all you require is a sufficient degree of sparsity that would um lead to the condition independences that are definitive of Markoff blankets and in one sense much of the sort of um the um the maths behind the free energy Principle as it inherits from random dynamical systems or uh in physics say Lan um equations is simply showing that a sparse

coupling a sparse causal coupling in a physics or control theoretic sense not a philosophical or Granger um or um cause effect structure sense but in the physics sense of a coupling so the the change um this state um influences the rate of change of that state so that kind of um which could be written as an influence graph so if there's a sparse coupling on the influence graph the equivalent basian graph or probabalistic graphical model induced by that sparse coupling um now uh can be interpreted in the in the sense of um um Pearl's Markoff blanking um so all you need to get to the application of the renormalization group to markof blankets is sparse causal coupling hence sparse Rand random dynamical system that is sparely coupled just out of interest um the spin block um the motivation for the spin block applying the spin block um or blocking Transformations that basically comes from the assumption that um many systems that people deal with both in computer vision and in uh and in um idealized um models in physics um has an exquisitely sparse distribution and that's because the causal inferences are all local so as soon as you in um you look at a a coupled map l ltis or a globally coupled map you'll notice let's take the ltis for example let's take a sort of idealized um uh um um Crystal uh so what you're saying when you write down a ltis is I'm only connected to my neighbors to my more neighborhood or to my Markoff blanket um that means that in a large latis uh where I could be connected to N^2 neighbors I'm only connected to 1 2 3 4 5 6 7 8 so a fantastic Al small almost empty set of possible connections and I use that phrase because there I can't remember who said it now but I really liked it the notion that if one looks at the connecto in the brain in terms of the connections it's almost empty you know the number of actual connections you know white matter connections external connections and indeed intrinsic um um exetral connections are almost zero in comparison to the total number of possible connections between the 10 to the 11 or whatever uh neurons in the brain so that's not surprising and I you it's one of those nice examples that the the anatomy the functional architecture of the brain recapitulates the anatomy of the lived World in it has this incredible sparcity um um and that just means um the universes in which there is um the action at a distance is quite rare um um means that that you you're just interacting you're just coupled to your neighbors and that's and that gives you if you then look at the mark of blanket structure or the particular partition for these lattice structures you get to the Sim block um blocking Transformations as a way of carving up nature you know in terms of little patches or Markoff blankets I do have a followup question on that but then I'll hand it back to Daniel or Lance um so the ltis example is really interesting because I think there's a symmetry there that maybe we lose when we talk about blankets that only have one parent so in a normal lce as you say

you're just connected to your eight Neighbors in 3D space when we looked at the spin block Transformations before we were going let's look at this square and everything in this square has one parent but the square along is connected to a separate parent so now if you're at the boundary we are breaking some symmetry there yes technically or practically a very good point um it doesn't really you it doesn't really um uh have any material impact upon the the the algorithms or the partitioning um so just practically what happens is you you you you you just take your tiles and then you have a half tile or a partial tile um until you get to the boundary so these these groups in the grouping operation or the or the blocking operation are not necessarily the same um you know they're not the same size um and indeed in uh um current um work we're actually dealing with sort of multiple streams so you can have sort of audio files and video files separately tiled and then converging right right at the top um ideally what you'd want to do is is um not use this um um blocking transformation but to try and identify the Markoff blanket structure at the particular scale in question actually use the Markoff blankets based upon um well there are a number of ways of doing that you can either use a markof blanket Discovery algorithm of the kind that Jeff Beck um Works upon um or you can use um empirical um analysis of the um the dynamical coupling um uh by estimating effective the jacobians you know for those people are interested there's a worked example of that in the particles and Parcels in the brain paper in network Neuroscience um it's really founded in terms of data analysis but you can use the same principles to um in to identify a particular partition from a large number of um um States or boxal or or or or or pixels um um you in three and 2D um respectively um there's another thing though that I think your question speaks to um which is um you know you've got you got you know one block of four and then another block of four and then um because of the sparcity that they're separately U modeled at at the level above so I thought you were going to ask how in Earth now do you start to repair the obviously rather facile assumption that you've got just local action because of course in our world there's lots of action to distance you we are um talking to each other over over over thousands of miles or Kil kilometers and you know we have Vision uh so all the and we have uh linguistic communication via sound so you know it is interesting that we have actually violated much of the local action if I was a virus this would be perfectly okay because the only thing that's really going to affect me as a virus is the things that actually are in molecular contact with me but for you and me um and the worlds in which we live in um we can't there is no such thing as just the action distance now starts to really matter um and effectively because the two um latent States generalized latent States generating stuff um initial

conditions and paths for the two blocks below because they themselves now constitute a single uh contribute to a single um latent State at the at the level above as you move deeper and deeper you start to now repair the simplifying Assumption of conditional Independence between the groups so it is at the higher levels that you now become you're able to see these action this action at a distance these coupling your what what happens over here really matters in terms of what's going on over here but you'll only ever see that you only have that as part of the generative model right at the top which is why all the inductive inference is implemented right at the top level where where you can see everything um so that it becomes you know if you like you are now you are now in a position to um um model and thereby generate appropriate predictions that rest upon um these long range dependencies um and of course you know us using the word long range I mean in the sense of the the coupling distance uh you you in a dynamical sense is that what you had in mind uh that was exactly what I had in mind thank you thank you on Lance do you have any remarks or want to add a question or I can yes please go for it yeah I have lots of questions uh thanks a lot C for the talk so I'm just wondering about this all roads lead to Rome slide that you had at the end and I'm wondering whether you have some sort of curse of dimensionality as you increase the dimensionality of the system so it sounds to me like the manifold of expert play because um because you're taking temporal trajectories that's going to be one dimensional manifold now as the state space of the game um increases in dimensionality you're going to have an ambient space of trajectory that's much much higher dimensional and so it sounds like the amount of trajectories they need to you know construe to bring me back to your um onedimensional expert manifold is just going to grow and grow and grow and it sounds like it might be hard to fill that space so could you could you comment a little bit on that and whether it's a problem right yeah in general in my world you know these things aren't problems but they're challenges um and I think that challenge is yet to be met to be met um so what I imagine people will do um is that you will um ingest um extra episodes or sequences that bring you onto your attracting set up until some bound um and then you will switch off active selection by which I mean the structure learning um so you stop growing the model and then you switch on parameter learning so that now you're forcing the Transitions and the um expression of various General Iz states to accommodate variations around the training data that you have been previously exposed to so that fixes the size of the tensors and hence the dimensionality of the state Bas State spaces in question um um but will in principle now accommodate by preserving by using sort of um Active Learning in the technical sense that you only update a parameter if it um

does not uh increase expected free energy or decrease Mutual information um then um you you're now in a position to become more robust um and perhaps I should qualified this answer the whole point of this adding in extra if you like insets onto an attracting set is just to equip the agent with a certain robustness should itself find itself outside or off that attracting set so um I know Lance knows this but I'll just make this explicit for for everybody else um this kind of learning um and state action policy um implementation is very much like learning to ride a bike um so you you basically um just accumulate and remember those experiences that were successful so for example I you know managed to um rotate the pedals by one cycle and I don't fall off and I remember that and then I another go and I wait until I've actually done two cycles and I don't fall off and I remember that I forget everything every time I I I fall off and then I keep on going until at some point I connect back my last cycle was identical to the first cycle and then I've learned to ride the bike but because I've forgotten all the falling off I can't recognize or know where I am when I when I've fallen off so I've got a very little expertise that will not allow me to recover if I fall off the edge so another example which makes it slightly clear I think is like mountain climbing you know if I want to be an expert mountain climber I cannot learn by my mistakes provided I'm climbing a sufficiently high mountain should I fall off it then I will die another example here is learning how to cross the road as a child you know you cannot learn by um um experiencing being run over but the cost you um you pay or the price that you pay is you can't recognize when you are being run over um and that I repeat lends a certain brittleness and a lack of robustness to agents who may be exposed to Danger that doesn't actually terminate them so putting this robustness back in by growing the um by growing the attracting set or basically putting these insets on um in this sort of generalized um phase space um you know it's just to make it more robust um in numerical studies it it does seem to um converge it doesn't seem to grow inde definitely you so those are likely violations are moving off the attracting set are themselves quite small relatively small in number um yeah practically speaking uh if if you've control the sources of stochasticity or unpredictability so it may not be always the case that you need to have an explicit bound on the upper the size of the um the very high level tensors if if you do then one has to now think of a principled way of making that upper bound an emergent property of um um expected for energy minimization it may be that you know you violate now the mutual information that certain rare events are just so rare you're actually compromising the mutual information by including them as a particular option um or a particular column of of a likely likelihood mapping or yeah but I don't know but these are all really interesting questions to

pursue thank you Carl all right Arun go for it um yeah so following on from that I think there an interesting question on the uniqueness part of ingesting data and the question is are there many more ways to fail than there are to succeed so if you're passing your data into a sort of Maxwell demon Maxwell's demon uh with a gate saying accept this data reject this data if this uh improves my Mutual information if there are many more ways to fail but then get back onto the a successful path but there are only a few paths that actually keep you there would it not make sense to just continually accumulate these sort of recovery States because the mutual information will keep increasing and then you stop there then you parameter you learn parameter wise because we're not including the sort of Prior distribution of How likely you are to be succeeding at any given moment because with the uniqueness thing you don't know oh 10 times out of 10 I'm cycling or like nine times out of 10 I'm cycling and one times out of 10 I'm falling off the bike yeah I think it depends um again I don't have any explicit numerical analyses or analytic analysis of this but intuitively um I think during learning there may well be um um good mileage in in gesting failures um or roots to success if you like or to the attracting set that that actually Traverse regimes that would normally be excluded by my constraints or my PRI preferences um you know just taking the view of the role of of the sea tensors or prior preferences from the point of view of physics what that basically is saying is it's actually carving out vast regimes of State space latent State space that you should not be in because the it would be uncharacteristic of you to be in these things so it it's really not so much about the rewards and the preferences uh it's really what you where you shouldn't be and that's what defines the attracting set um however before you've learned that attracting set it may be useful to include constrained regions of State space that offer a root into or onto that attracting set so I can certainly see that that would be a useful device but notice that once you're on the attracting set you will never visit the constrained regions because you now keep yourself on the attracting set um um and therefore if you've encoded those but you never visit them your Mutual information will actually fall so what that means operationally is what you'll probably do is have this sort of slightly over inclusive structure learning learn how to cope with all the necessary sources of Randomness so that your attracting set is sufficiently robust and you do actually visit from time to time but not um not never um s of excursions from the attracting set are drawn back onto the attracting set um and then use basing model reduction to eliminate all the states that you don't want to recognize simply because you never go there um you know once in a you know when I was younger I would certainly be able to rec what it's like to fall off my bike but now as an expert bike rider I really don't want or need

that so you know and that's just basically getting old and wise and you know implementing bisecting model reduction to remove the Redundant parameters which would in this instance just be the columns u_m and u_m Associated slices or or rows of the of the transition matrices so I think you could perhaps another way of then ADV answering Lan's question is yeah you just keep u_m accumulating u_m but u_m start to engage Bas in model reduction at some point you should reach an equilibrium which will be the attracting set that allows you to survive in the character states that Define you in this particular environment with this particular volatility thank you awesome okay I'm going to bring in a question from the live chat okay Javier wrote so stochastic plus dynamical probabilities learned by the agent itself are scale-free will this hierarchical learning also be applied to the goals of the agent different abstractions granularities of goals [Music] self-learned my answer would be yes but I'm going to I'm going to ask Lance to to to answer that one oh that that's cheeky uh yeah I would say for sure u_m because typically when we have when we oper operational operationalize sorry the goals in our gen model they need to have the same structure at our Gena model itself so I would imagine then we would have a hierarchical renormalizable goals which kind of makes sense cool okay I'll bring in oh yeah go for it Carl I was just going to say u_m you know just to u_m unpack what Lance just said in terms of practical things you know if you're trying to u_m you're trying to say u_m what my goal is is to go uh is to have dinner u_m that naturally unpacks hierarchically into I've got to move from the living room to the kitchen then I have to find the you open the fridge then sorry I have to find the food which then unpacks into I have now open them and all the way down to to the finest finest muscle movement so almost by definition u_m your know intended narratives can always be u_m decomposed in this scale-free way you u_m there there's a this is very similar to sort of u_m motor chunking and hierarchical decomposition in in u_m u_m theoretical uh motor control for example and you probably find it a lot in in robotics there's a global long-term plan that that provides empirical priz of constraints on shorter little chunks that themselves provide constraints on shorter and shorter chunks and that's exactly the u_m if you like the the architecture implicit in these scale free models awesome and totally ties back to the synergetics okay I'll ask two questions from ml Dawn he wrote how can we inject the same separation of time scales by imposing smoothness constraints on say the parameters as opposed to playing with proportional update rates at different levels and this relates to something Arun and I talked a lot about in the zero and leading up to it is identifying which Windows to use you could imagine picking the right separation of time scales if the classes on campus really do shift on the hour then the hour

course graining is going to be great and effective but the 59 or the 61 minute might Alias you with the real variability of the system and so there's not necessarily even a smooth landscape for determining the windowing or the binning of coar graining and coar graining may have dependences on each other so how do you go about playing and finding viable approaches and robust ones amidst all these different levels of of meta optimization that are opening up I'll take this one can take take the next one um then that's a really good question because there is an implicit separation of time scales between um so vanilla implementations of inference and learning however in these schemes there isn't um so the the know the um the Matlab version or implementation um of um active inflence under these and the fact all uh markof decision process models um unlike um the um procedures that you might find in data analysis where you acquire you basically do some filtering on the on a an Epoch of data and then at the end you update your parameters and then you rinse wash and repeat um in um in active inference um simulations and also in some um applications of what's called generalized filtering you have continual learning so the update of the parameters actually occurs a at every time step um at each level of the model so the likelihood parameters at the lowest level are updated every time that routine is called um so there is continual learning the learning um is yes you're absolutely right is smooth in the sense that you're slowly accumulating drish parameters but the tempal scale of that learning is the same as the inference um and you could also argue well could one also put model reduction um into play at the same time scale so basically can you at every time point at each level in the model um identify redundant parameters and remove them uh using Bas and model reduction in effect that's what's done uh using Active Learning what I said before Active Learning will only proceed um if the expected free energy of mutual information improves that's actually also implemented so the the kind the the the rate the rate at which it looks as though things change um between inference learning and model selection is certainly distinct I don't think that's quite the same kind of separation of time scales that is implied in the renormalization group that's as you were intimating uh Daniel exactly specified by the number of ticks or updates at level n per single update date at the uh level $n+1$ um and that is I think you the you the key question you're asking how do you get that right um and of course the answer to that is the model that has the greatest model evidence or smallest free energy or greatest marginal likelihood for the kind of data generated by this environment in this world so um are there any um sort of useful prizes you can can put over this um well what one one useful prior is um that um things are changing as uh to to well one PRI which is actually used in operation is you take the lower limit on on What's called the the temporal RG

flow which is basically the the length of the window at each and every level Δt at the moment that's two what does that mean how can you motivate that well it's the most expressive in the following sense but if I Δt just imagine Δt what imagine what this discrete discretization of time means from the point of view of classical Notions in continuous State space like position and momentum or position and velocity if Δt if I use a Δt a scaling of two which is basically means there are two time steps or every time step the Δt the level above what I effectively doing is Δt modeling two time steps for every state at the level above and that basically means I can model the initial condition and the subsequent condition basically terms of the change which of course is the velocity Δt and what does that mean for the level above that well it now means that I'm modeling the change in the velocity so I'm now modeling the acceleration and then so on up until Δt you know the high order of motion so what that means is by taking this limiting case of Δt the temp temporal horizontal the the you the ratio of time time steps from one level to the other level Δt as two means that I got a very expressive model that can handle Δt can recognize Δt things that change very very quickly and their acceleration their jerk and all and the implicit Δt context sensitivity that that that gives you if I didn't do that I'd have to assume that I C grain over long periods of time and therefore the path is consistent during that time now that sometimes is a useful approximation and indeed there's a whole industry called you know called linear switching linear dynamical systems that makes that approximation it means you can Δt approximate all the nonlinear complicated high order motions in some evolving pattern or system with a series of Δt linear trajectories that basically go you know that intercede between a series of points at which you which you switch the problem with that is that the length of the linear trajectories for as for example I'm committed to one Path Δt so one slice of my B Matrix for the next next eight time steps the problem with this is that you don't know when you when you deviate from that transition Dynamics so then you get into the game or perhaps I could make my scheme reactive so sometimes I do three steps sometimes I do eight steps sometimes I do one step and I have some criteria for saying your predictions about my trajectory have now been met please Δt I'm going to tell you give you the evidence that I've been pursuing this path you up dat your priv level above and tell me what you expect me to do next so this now speaks to Δt a particular kind of reactive message passing Δt that starts to have implications for the computer science and the sort of CR the crud Δt or Δt as Δt as I think we've rehearsed before Δt The Institute the the original conception uh behind Δt the actor model in computer science from the 1990s so it's all about reactive message passing I will send you a message

when you send me I will return a message to you if you send me a message um and what this um what these um renormalizing time models have and I should say that this also applies in the absence of the the spatial renormalization so any deep temporal model will have this aspect what that means is that from the point of view of Any Given level I will um be responding to messages more quickly from my subordinates than from my superiors so once in a while I'll get a request from a superior my response to that is to ask my juniors to do a series of things they report back to me and when they report I give them the next bit bit um request or guidance uh until they complete their job and when all my juniors have completed their job of course they be normalizing specially normalizing context I'll have a whole group of of Juniors to worry about um then um I have gathered enough evidence for me to then return the the message to my to my Superior so you get the in terms of the reactive message passing in this sort of actor model of um of Compu to message passing um you get the the separation of temporal scales inherits from the number of messages or re um requests and uh responses um um exchanged with the people lower down in the hierarchy relative to the number immediately subordinate um and in principle they don't have to be fixed so there could be a criteria um before I pass the message back back up to the to the higher level I think that's Daniel probably what you were getting at um I'm not there may be somebody who's done that um um but I'm I haven't done that and I I I don't know uh I can't give you an example of somebody who has done that you may find you may find somebody um in Bert de's group um who sort of you know some of the world experts um in reactive message passing who have considered these kinds of things um you know you in in principle um you know there should be a way of um each level testing whether I have fulfilled the predictions of my superiors um and now it's time to get more guidance you know uh based upon the energy awesome the organizational technology I did didn't know that my PhD needed okay I'll ask a question as we get close to the end and and this will be for for anyone ml Dawn wrote how does the training time of active inference compared to that of reinforcement learning and how does their scalability compare and it would be interesting just to hear anyone's very 2024 takes where are we at with that entire Arc from problem conceptualization on through the training and the inference how do we even grapple and describe with similarities and differences in the training time and resources and all of this that's definitely Lance for for at least five minutes I think I'm I'm a bit incapacitated in terms of talking um I think this is actually a good question for Aron um but yeah I mean I can say a few things um so uh scaleability in actor inference I would say it has been a challenge and there has been um much less man work hours that have been put

into that than in reinforcement learning now for the past few years we we've had a lot of work um a lot of it led by C actually um trying to scale the structure learning in active inference and and I think we've done a lot of mileage on that so there's been the supervised structure learning paper there's this paper uh there was also some interesting work um presented at the international work shop and active inference um I personally think there are still some uh challenges ahead and and I think uh one of them is is the one that we just talked um a few questions ago about you know if you if you learn an expert orbit how do you get back on the orbit as the dimensionality of the system increases so this is definitely um one challenge um now I know much less about reinforcement learning so um yeah maybe Arun has more things to say there um but but what I will say is a lot of reinforcement learning is powered by Deep neural networks when we use deep neural networks we sacrifice uh the interpretability of our generative model this is one of the strength of active inference is having more interpretable um Dynamic Cal models where we can actually peer into them and see okay well this representation includes this this other one include that and so forth um the fact though that reinforcement learning typically uses deep neural networks means that it has been much more scalable uh just because there has been so much work done in like training deep networks this is a bit of my uh my perspective I don't think it fully answers the question though I can come in with a little bit on that but I must admit that I've not trained any reinforcement learning agents myself I worked with people who have uh I'd agree Lance that the infrastructure generally is much more advanced for training RL agents for doing stuff uh and completely agree with as you say the explainability and the interpretability of why did an RL agent pick a particular action is very opaque uh and I think the deeper those networks become the harder it is to understand what's going on in the hidden layers and uh generally why they do what they do I think with Act of inference yes we do have more explainability but I do wonder if we go the far structure learning approach I think Carl had a particular uh comment earlier about looking at the highest level um identifying the desirable States learned by the agent from what I understand the identification has to be done by a human they have to look at the model that's been learned by this uh renormalization approach look at those top levels and then say ah what do these levels mean so an example in the paper is the Mist uh classification it's the case of a human being looking at those top levels and going ah these top levels each correspond to a digit in the handwriting task so we are uh sacrificing some explainability I think by doing this fast structure learning compared to what might be the old fashioned way way of creating uh G models By Hand by thinking sitting in my armchair and thinking this is how I might model the

world these structures these likelihoods these states and when you do that I think just by hand you have 100% explainability that model just might not perform very well so I think there's always going to be that tension there um but in terms of how quickly these things learn um having not trained either I'd be having a reckon and at the moment I would say probably an RG um active inference agent would train faster simply because it's able to use the structure in the data much more effectively that would that would be my take and I think as well really interesting thinking about the active data selection part of this too I know that there is some stuff in the paper about the efficiency being both a curse and a burden but I'll hand over to Lance and Carl to give more details on that yeah Lance yeah just to um echo echo what Arun said um also so so the structure learning active inference and in reinforcement learning is is also very different um in reinforcement learning we typically have a a model with deep n networks that's massively overparameterized but it has some fixed representational capacity from the outside because the structure of that of that model will be specified and then typically the model will learn and then probably you would prune it using some techniques like Dropout or or other things if we looked at that from the perspective of active inference it would be like specifying a massively overparameterized model and then pruning it with Bas model reduction the um but but now in active inference we have this Bas and model expansion for example in this paper and what this Basin model expansion does is it learns from the get go the minimal model for explaining the data and so in that sense it's extremely efficient and also provably because it extremizes the um the marginal likelihood or the mutual information yeah I just wanted to Echo that last Point that's a really important Point um so Lance is drawing the distinction and there are many distinctions you can draw with um deep RL um and um deep active inference read in a more in in its um generic sense um and that that distinction at the level of structure learning or model selection I I think is absolutely crucial so you know deep RL starts with an overly expressive model with a known functional form um that is deliberately overparameterized and then shrinks it or reduces it to um maximize the mutual information the expected free energy sometimes they um referred to as a cross entropy in machine learning so the reward function now becomes uh there's no explicit reward in unsupervised learning it's just replaced by the mutual information or the cross entropy um which should be distinguished between supervised reinforcement learning such as you know digit classification for example um whereas the you know active inference um under these re under fast structure learning is exactly as Lance says it starts from nothing and stops when it's minimally complex which means it's maximally efficient and that efficiency translates in many

guises through to sample efficiency through to um um statistical efficiency through to thermodynamic efficiency and this matters because from the point of view of climate change this is the right way to do it as opposed to building power stations to power large language models and Transformers that need big data so um the efficiency um on all levels whether it's the the sample efficiency or using the right data uh or whether it's right through to how much electricity do you need to actually train your model and how many millions of dollars do you need to trade a foundation model um all of these are reflections of the failure of um machine learning to write in the efficiency into the ultimate objective function which is un said is just the marginal likelihood of the evidence for this model of of this kind of content or or World awesome okay final question and a very apt one as we head into the rest of the Symposium so just give any short thoughts Tintin wrote hi all thanks for the incredible event does the sequence does the sequence in which the training data is fed to the A structure learning model impact what the resulting learn structure would be so the learning curriculum AKA Symposium program challenge well the answer is sorry I'm mindful that Lance is recovering from his um jaw surgery um yeah absolutely um and this is um again just in relation to to to to the last question something I think that eludes uh conventional RL certainly that it rests upon um um classification procedures and um because um the active inference as an application the free energy principle is all about the Dynamics of self-organization and Dynamics means time and time means sequence matters so inevitably at some level the order in which you see things is absolutely crucial for getting the right kind of generative model um that is even in the case um actually of static image recognition because you're assuming that there's no ordinal structure or correlation structure to the presentation say of emus digits you're assuming that they are selected at random as opposed to seeing all the tens and then all the all the ones and then all the all the sevens um so even you in these edge cases um um which are you know the focus of um 20th century RL um you know static image classification for example um your order does matter but they they it matters absolutely acutely because of course this the order in which you um see um content in um the infast structure learning determines how you populate the um the transition tensors and it's a transition tensors that enable the separation of temporal scales that enables the ability to um encode narratives and plans that far transcend the actual rate at which you samp sample the environment you could also argue that you know this issue has emerged um in a in a strange kind of guys in Transformer models so the attention heads operate on the past so the the where you select the from the past in terms of which token is going to predict the next token most efficiently um again is exquisitely dependent upon and

sensitive to the order in which you train train train these models So Daniel have you thought carefully about your ordinal approach today I I let the structure I let the structure of the niche and others' constraints provide the first pass and make requests from there so thank you Carl thank you Lance for joining again thank you run for all the contributions with azero Carl great to see you so next up we'll take a 30 second break and we'll be back with Michael Lenin so thanks fellows see you soon e